Got it 👍 Let's go point by point in **simple but detailed exam-ready notes** for you:

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### 1. Comparison between Soft and Hard Computing

| Feature | Hard Computing | Soft Computing |
|---|---|---|
| Definition | Traditional computing, based on binary logic and exact models | A branch of computing that deals with approximate models and uncertainty |
| Basis | Uses **crisp logic** (0 or 1, true or false) | Uses **fuzzy logic, neural networks, genetic algorithms** |
| Approach | Deterministic (exact answer) | Probabilistic/approximate (good enough answer) |
| Flexibility | Rigid, strict rules | Flexible, can handle imprecision |
| Example | Arithmetic operations, database queries | Weather forecasting, handwriting recognition |

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### 2. Applications

🔹 **Hard Computing Applications**

- Banking transactions (ATM PIN validation)
- Inventory management
- Compiler design
- Exact arithmetic calculations

🔹 **Soft Computing Applications**

- Weather prediction
- Image & speech recognition
- Medical diagnosis (handling uncertain symptoms)
- Stock market analysis
- Control systems (washing machines, air conditioners with fuzzy logic)

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### 3. Difference between Crisp and Fuzzy Logic

| Aspect | Crisp Logic | Fuzzy Logic |
|---|---|---|
| Truth Values | Either **0 or 1** | Any value between **0 and 1** |
| Nature | Rigid, black & white | Approximate, shades of gray |
| Example | "Temperature is Hot" → If >30°C = True, else False | "Temperature is Hot" → 28°C = 0.6 hot, 35°C = 0.9 hot |
| Used in | Hard computing | Soft computing |

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**4. Crisp Set and Fuzzy Set**

- **Crisp Set** → A set with well-defined boundaries.
  Example: Set of even numbers {2,4,6,…}
  Either an element belongs or does not.
- **Fuzzy Set** → A set where elements have **degrees of membership**.
  Example: Set of "tall people" –
  - Person A (180 cm) → 0.9 membership
  - Person B (160 cm) → 0.4 membership

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**5. Crisp Relation and Fuzzy Relation with Example**

- **Crisp Relation**: A binary relation between sets, defined in a strict (yes/no) manner.
  Example: Relation "is greater than" on {1,2,3} → {(2,1), (3,1), (3,2)}
- **Fuzzy Relation**: Assigns a **membership value** (between 0 and 1) to each pair.
  Example: Relation "is approximately equal" on {2,3,4}
  - (2,3) → 0.7
  - (3,4) → 0.8
  - (2,4) → 0.2

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**6. Operations on Fuzzy Set**

1. **Union (A ∪ B)** → $\mu_{A \cup B}(x) = \max[\mu_A(x), \mu_B(x)]$
2. **Intersection (A ∩ B)** → $\mu_{A \cap B}(x) = \min[\mu_A(x), \mu_B(x)]$
3. **Complement (A′)** → $\mu_{A'}(x) = 1 - \mu_A(x)$
4. **Difference (A − B)** → $\mu_{A-B}(x) = \min[\mu_A(x), 1 - \mu_B(x)]$
5. **Algebraic sum/product** (advanced operations for soft computing)

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**7. Fuzzification (also written "Fuzzyfication")**

- **Definition**: The process of converting **crisp inputs (exact values)** into **fuzzy values (degrees of membership)**.

**Methods:**

1. **Singleton fuzzification** → Crisp input maps to a membership value directly.
2. **Gaussian fuzzification** → Uses Gaussian membership function for smooth mapping.
3. **Triangular/Trapezoidal fuzzification** → Uses simple shapes for mapping inputs.

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**8. Fuzzy Inference System (FIS)**

- A framework that applies **fuzzy logic rules** to make decisions.

- **Steps**:
    1. Fuzzification (convert crisp input to fuzzy)
    2. Rule Evaluation (apply fuzzy rules: IF-THEN)
    3. Aggregation (combine results of all rules)
    4. Defuzzification (convert fuzzy output back to crisp value)

**Types**:

- **Mamdani FIS** (common, uses linguistic rules)
- **Sugeno FIS** (used in control systems, output is function-based)

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**9. Membership Function Calculation (in detail)**

- A **membership function (MF)** defines how each element maps to a degree of belonging (0–1).

**Common Membership Functions:**

1. **Triangular MF**

   μ(x)={0x≤a or x≥cx−ab−aa<x≤bc−xc−bb<x<cμ(x) = \begin{cases} 0 & x ≤ a \text{ or } x ≥ c \\ \frac{x-a}{b-a} & a < x ≤ b \\ \frac{c-x}{c-b} & b < x < c \end{cases}

   Example: For “medium temperature”, a=20, b=30, c=40

2. **Trapezoidal MF**
   Similar to triangular but has flat top.
3. **Gaussian MF**

   μ(x)=e−(x−c)22σ2μ(x) = e^{-\frac{(x-c)^2}{2σ^2}}

   Smooth curve centered at **c** with spread **σ**.

4. **Sigmoid MF**

   μ(x)=11+e−a(x−c)μ(x) = \frac{1}{1+e^{-a(x-c)}}

- Membership values are calculated by substituting **x (crisp input)** in these functions.

Perfect 👍 Vanshika! Let's expand everything in **exam-style detailed notes with formulas + explanation**.

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# 📘 Detailed Notes on Soft Computing & Fuzzy Logic

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## 1. Comparison between Soft and Hard Computing

| Aspect | Hard Computing | Soft Computing |
|---|---|---|
| **Definition** | Based on deterministic models, exact logic | Deals with approximate reasoning and uncertainty |
| **Logic Used** | **Crisp logic** (binary, 0 or 1) | **Fuzzy logic, probabilistic, neural nets** |
| **Approach** | Algorithmic, exact steps | Heuristic, adaptive, tolerant of imprecision |
| **Accuracy** | Always exact | Approximate but acceptable |
| **Flexibility** | Rigid | Flexible, can adapt |
| **Examples** | Calculator, compilers, database queries | Weather prediction, handwriting recognition |

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## 2. Applications

🔹 **Hard Computing Applications**

- Banking systems (PIN verification, transactions)
- Compiler design (exact parsing)
- Railway reservation systems
- Digital circuits (0/1)

🔹 **Soft Computing Applications**

- Pattern recognition (fingerprint, face recognition)
- Natural Language Processing (NLP)
- Medical diagnosis (diseases with uncertain symptoms)
- Stock market forecasting
- Smart appliances (AC, washing machine fuzzy controllers)

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## 3. Difference between Crisp and Fuzzy Logic

| Aspect | Crisp Logic | Fuzzy Logic |
|---|---|---|
| Truth Values | Only **0 (false)** or **1 (true)** | Any value in **[0,1]** |
| Nature | Rigid, black & white | Approximate, shades of gray |
| Example | "Temperature > 30°C → Hot" → strictly true/false | "Temperature is Hot" → 28°C = 0.6 hot, 35°C = 0.9 hot |
| Used In | Hard computing | Soft computing |

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# 4. Crisp Set and Fuzzy Set

- **Crisp Set**:
  Membership function:

  μA(x)={1,x∈A0,x∉Aμ_A(x) = \begin{cases} 1, & x \in A \\ 0, & x \notin A \end{cases}

  Example: A = {even numbers} → μ(2)=1, μ(3)=0

- **Fuzzy Set**:
  Membership function:

  μA:X→[0,1]μ_A : X \to [0,1]

  Each element has partial membership.

  Example: A = {“Tall people”}

  - μ(180 cm)=0.9
  - μ(160 cm)=0.4

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# 5. Crisp Relation vs Fuzzy Relation

- **Crisp Relation (R)**: Subset of Cartesian product.

  R⊆A×BR \subseteq A \times B

  Example: A={1,2,3}, relation “>”:
  R = {(2,1), (3,1), (3,2)}

- **Fuzzy Relation (Ṙ)**: Membership function assigns a degree.

  μR:A×B→[0,1]μ_R : A \times B \to [0,1]

  Example: A={2,3,4}, relation “≈” (approximately equal):

- $\mu(2,3)=0.7$, $\mu(3,4)=0.8$, $\mu(2,4)=0.2$

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# 6. Operations on Fuzzy Sets

Let $\mu A(x), \mu B(x) \in [0,1]$

1. **Union**:

   $\mu A \cup B(x) = \max(\mu A(x), \mu B(x))$

2. **Intersection**:

   $\mu A \cap B(x) = \min(\mu A(x), \mu B(x))$

3. **Complement**:

   $\mu A'(x) = 1 - \mu A(x)$

4. **Difference**:

   $\mu A - B(x) = \min(\mu A(x), 1 - \mu B(x))$

5. **Algebraic Product**:

   $\mu A \cdot B(x) = \mu A(x) \cdot \mu B(x)$

6. **Algebraic Sum**:

   $\mu A \oplus B(x) = \mu A(x) + \mu B(x) - \mu A(x) \cdot \mu B(x)$

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# 7. Fuzzification

**Definition**: Conversion of **crisp input (numerical)** into **fuzzy values (linguistic terms)**.

## Methods:

1. **Singleton fuzzification**
   - Input is matched with single membership value.
2. **Triangular fuzzification**
   - Uses triangular membership function.
3. **Trapezoidal fuzzification**
   - Flat-topped fuzzy mapping.
4. **Gaussian fuzzification**
   - Smooth bell curve-based mapping.

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# 8. Fuzzy Inference System (FIS)

**Steps:**

1. **Fuzzification** → Convert crisp input to fuzzy values
2. **Rule Evaluation** → Apply rules (IF–THEN)
   - Example: IF temp is high AND humidity is medium THEN fan speed is fast
3. **Aggregation** → Combine results of all rules
4. **Defuzzification** → Convert fuzzy output → crisp value

**Types of FIS:**

- **Mamdani Model** (linguistic output)
- **Sugeno Model** (mathematical output functions)

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# 9. Membership Function (MF) Calculation

MF defines **degree of membership** (0–1).

## Types with Formula

1. **Triangular MF**

   μ(x)={0x≤a or x≥cx−ab−aa<x≤bc−xc−bb<x<cμ(x) = \begin{cases} 0 & x \leq a \ \text{or}\ x \geq c \\ \frac{x-a}{b-a} & a < x \leq b \\ \frac{c-x}{c-b} & b < x < c \end{cases}

2. **Trapezoidal MF**

   μ(x)={0x≤a or x≥dx−ab−aa<x≤b1b<x≤cd−xd−cc<x<dμ(x) = \begin{cases} 0 & x \leq a \ \text{or}\ x \geq d \\ \frac{x-a}{b-a} & a < x \leq b \\ 1 & b < x \leq c \\ \frac{d-x}{d-c} & c < x < d \end{cases}

3. **Gaussian MF**

   μ(x)=e−(x−c)22σ2μ(x) = e^{-\frac{(x-c)^2}{2σ^2}}

4. **Sigmoid MF**

   μ(x)=11+e−a(x−c)μ(x) = \frac{1}{1+e^{-a(x-c)}}

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Great, Vanshika 👍 Let's build **complete exam-ready notes** on your given Soft Computing topics.

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# 📘 Soft Computing – Complete Notes

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## 1. Introduction to Soft Computing

- **Soft Computing** is a branch of computer science that deals with **imprecision, uncertainty, approximation, and partial truth**.
- Unlike **hard computing** (traditional), which requires exactness, soft computing **mimics human reasoning** and provides **good-enough solutions** for real-life problems.
- It is inspired by biological systems, human reasoning, and natural evolution.

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## 2. Historical Development

- **1965 → Lotfi A. Zadeh** introduced *Fuzzy Set Theory* → foundation of fuzzy logic.
- **1970s–1980s** → Artificial Neural Networks (ANN) became popular for pattern recognition.
- **1975** → John Holland introduced *Genetic Algorithms* (GA) based on natural evolution.
- **1990s** → Emergence of the term **"Soft Computing"** (Zadeh).
- **2000s onwards** → Integration of fuzzy logic, ANN, GA, and probabilistic reasoning → applied in AI, robotics, medicine, finance.

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## 3. Definitions

- **Soft Computing**:
  "Soft Computing is an approach to computing that deals with approximate reasoning, imprecision, uncertainty, and partial truth to provide solutions that are cheaper, faster, and more robust."
- **Hard Computing**:
  "Hard Computing is a traditional computing approach that requires precise input, exact models, and deterministic solutions."

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## 4. Advantages of Soft Computing

✅ Can handle **uncertain, vague, or incomplete data**
✅ Provides **low-cost and robust solutions**
✅ Mimics **human-like reasoning**
✅ Applicable to **complex real-world problems** (where hard computing fails)
✅ Adaptive and tolerant to imprecision

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## 5. Disadvantages of Soft Computing

❌ Solutions are **approximate, not exact**
❌ Requires more **computational resources** (for ANN, GA)
❌ **Designing fuzzy rules** and tuning membership functions is not always easy
❌ Less effective for **precise numerical problems** (e.g., arithmetic, banking transactions)

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## 6. Solution of Complex Real-Life Problems

Soft computing is used where hard computing cannot give efficient solutions:

- **Medical field** → Diagnosis of diseases with uncertain symptoms
- **Engineering** → Adaptive controllers in machines and robotics
- **Finance** → Stock market prediction
- **Transportation** → Traffic signal optimization, self-driving cars
- **Weather prediction** → Handling chaotic, uncertain climate data

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## 7. Soft Computing and Its Techniques

Soft computing is a **hybrid approach**, combining several techniques:

1. **Fuzzy Logic (FL)** – deals with uncertainty and approximate reasoning
    - Example: Washing machines with fuzzy controllers.
2. **Artificial Neural Networks (ANN)** – inspired by biological neurons, used for learning & pattern recognition
    - Example: Face recognition.
3. **Genetic Algorithms (GA)** – optimization technique based on natural selection
    - Example: Route optimization in delivery services.
4. **Probabilistic Reasoning** – deals with randomness and uncertainty
    - Example: Spam filtering in emails.
5. **Hybrid Systems** – combination of above methods for better efficiency
    - Example: Neuro-fuzzy systems.

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# 8. Soft Computing vs Hard Computing

| Aspect | Hard Computing | Soft Computing |
|---|---|---|
| **Logic** | Crisp logic (0 or 1) | Fuzzy logic (0–1 range) |
| **Approach** | Deterministic | Probabilistic/heuristic |
| **Data Requirement** | Precise & complete data | Incomplete, uncertain data allowed |
| **Flexibility** | Rigid | Flexible & adaptive |
| **Output** | Exact | Approximate (close to real) |
| **Example** | Banking transaction, compiler | Weather forecasting, speech recognition |

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# 9. Applications of Soft Computing in Current Industry

Soft computing is widely used in **AI, machine learning, automation, and decision-making**:

- **Medical Diagnosis** → Cancer detection, heart disease prediction using ANN + fuzzy logic.
- **Automobile Industry** → Anti-lock braking system (ABS), automatic gear shifting using fuzzy controllers.
- **Consumer Electronics** → Smart washing machines, AC temperature control (fuzzy logic).
- **Finance** → Credit scoring, fraud detection, stock prediction.
- **IT & Data Science** → Natural Language Processing (chatbots, sentiment analysis).
- **Robotics & AI** → Autonomous vehicles, intelligent robots.
- **Agriculture** → Crop yield prediction, smart irrigation.

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Perfect Vanshika 👍 Let's prepare **detailed exam-ready notes** for your Fuzzy Logic topic with **formulas, explanations, and methods**.

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# 📘 Fuzzy Logic – Complete Notes

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## 1. Introduction to Fuzzy Logic

- Introduced by **Lotfi A. Zadeh (1965)** with the concept of *Fuzzy Sets*.
- Unlike **classical logic (crisp)**, fuzzy logic deals with **approximate reasoning**.
- It allows **truth values between 0 and 1**, representing *degrees of truth*.
- Example:
  - Crisp → "Temperature is hot" → (30°C = True, 29°C = False).
  - Fuzzy → "Temperature is hot" → 29°C = 0.7, 30°C = 0.9.

**Applications** → Washing machines, medical diagnosis, robotics, weather forecasting.

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## 2. Crisp Sets

- In classical set theory, elements either belong or do not belong to a set.
- **Membership function**:

  μA(x)={1,x∈A0,x∉Aμ_A(x) = \begin{cases} 1, & x \in A \\ 0, & x \notin A \end{cases}

- Example: A = {even numbers}, μ(2)=1, μ(3)=0.

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## 3. Fuzzy Sets

- A fuzzy set allows **partial membership** (between 0 and 1).
- **Membership function**:

  μA:X→[0,1]μ_A : X \to [0,1]

- Example: A = {tall people}
  - μ(160 cm)=0.4, μ(170 cm)=0.6, μ(180 cm)=0.9.

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# 4. Fuzzy Relations

- A **fuzzy relation** generalizes classical relations by assigning a **membership degree** to each pair.

**Mathematical Representation**:

μR:A×B→[0,1]μ_R : A \times B \to [0,1]

- Example: Relation "is approximately equal" on {2,3,4}
  - μ(2,3)=0.7, μ(3,4)=0.8, μ(2,4)=0.2.

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# 5. Membership Functions and Features

A **Membership Function (MF)** defines how each element maps to a membership degree.

## Common Types of MF:

1. **Triangular MF**

   μ(x)={0x≤a or x≥cx−ab−aa<x≤bc−xc−bb<x<cμ(x) = \begin{cases} 0 & x \leq a \ \text{or}\ x \geq c \\ \frac{x-a}{b-a} & a < x \leq b \\ \frac{c-x}{c-b} & b < x < c \end{cases}

2. **Trapezoidal MF**

   μ(x)={0x≤a or x≥dx−ab−aa<x≤b1b<x≤cd−xd−cc<x<dμ(x) = \begin{cases} 0 & x \leq a \ \text{or}\ x \geq d \\ \frac{x-a}{b-a} & a < x \leq b \\ 1 & b < x \leq c \\ \frac{d-x}{d-c} & c < x < d \end{cases}

3. **Gaussian MF**

   μ(x)=e−(x−c)22σ2μ(x) = e^{-\frac{(x-c)^2}{2σ^2}}

4. **Sigmoid MF**

   μ(x)=11+e−a(x−c)μ(x) = \frac{1}{1+e^{-a(x-c)}}

**Features of Membership Functions**:

- Normality → max μ(x) = 1
- Convexity → one peak (for simple sets)
- Continuity → smooth curves preferred

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# 6. Fuzzification

- Process of **converting crisp input** (numerical data) into fuzzy values using membership functions.

### Methods of Membership Value Assignment

1. **Intuition/Expert Knowledge** – Based on human expertise.
2. **Histogram method** – Uses data distribution to define membership.
3. **Clustering (Fuzzy C-means)** – Groups data and assigns memberships.
4. **Functional Method** – Uses predefined functions (triangular, Gaussian, trapezoidal).

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# 7. Defuzzification

- Converts **fuzzy output → crisp output**.
- Needed because systems require numerical values for control.

### Common Methods:

1. **Centroid (Center of Gravity)**

   x∗=∫x μ(x) dx∫μ(x) dxx^* = \frac{\int x \, μ(x) \, dx}{\int μ(x) \, dx}

2. **Bisector Method** – Divides fuzzy set into two equal areas.
3. **Mean of Maximum (MOM)** – Takes average of max membership values.
4. **Smallest/Largest of Maximum (SOM/LOM)** – Takes min or max x where μ(x) is maximum.

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# 8. λ-cuts (Alpha-cuts)

- A technique to convert a fuzzy set into a **crisp set** at a chosen threshold **λ (alpha)**.

**Definition**:

Aα={x∈X | μA(x)≥α}A_\alpha = \{x \in X \ | \ μ_A(x) \geq α \}

- Example: If fuzzy set "Tall" has values
  - μ(160)=0.4, μ(170)=0.6, μ(180)=0.9
  - For α=0.6 → A_α = {170, 180}.

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# 9. Fuzzy Measure

- A generalization of probability measure.
- A **fuzzy measure** assigns a real number to a set, but not necessarily additive.

**Properties**:

1. $\mu(\emptyset)=0$, $\mu(X)=1$
2. If $A \subseteq B \rightarrow \mu(A) \leq \mu(B)$ (monotonicity)
3. Non-additivity allowed (unlike probability).

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# 10. Fuzzy Reasoning

- Process of making decisions using fuzzy logic rules.
- **Rule Form**:
  - IF (condition) THEN (conclusion).
  - Example: IF temperature is high AND humidity is medium THEN fan speed is fast.
- Uses fuzzy operators: **AND → min, OR → max, NOT → 1–μ(x)**.

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# 11. Fuzzy Inference System (FIS)

A framework that applies fuzzy rules to map input → output.

### Steps in FIS:

1. **Fuzzification** (convert crisp input → fuzzy sets)
2. **Rule Evaluation** (apply IF-THEN rules)
3. **Aggregation** (combine outputs of all rules)
4. **Defuzzification** (final crisp output)

### Types:

- **Mamdani FIS** – widely used, linguistic output.
- **Sugeno FIS** – output is mathematical function, good for control systems.